THE DEPTH AS THE SMALLEST DEGREE IN WHICH A NORMALIZED DUALIZING COMPLEX IS NONZERO

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The degree confinement for the cohomology of a normalized dualizing complex against a finite module. Invoked at the start of the counted proof (source line 3817). Gives the result directly, and is the reference point for the two alternative arguments.
- The duality statement identifying Ext of a module into the dualizing complex with Ext of the residue field into the ring. Invoked inside the first alternative (source lines 3827-3829). Rewrites the smallest nonvanishing degree as the smallest degree in which Ext of the residue field into the ring is nonzero.
- The characterization of the depth as the smallest degree in which Ext of the residue field into the ring is nonzero. Invoked inside the first alternative (source lines 3830-3831). Identifies that smallest degree with the depth.
- The vanishing criterion for local cohomology in degree equal to the depth. Invoked inside the second alternative (source lines 3836-3838). Gives the second derivation of the statement through local duality.

Source reference. Original reference algebra-lemma-depth-ext: sources/stacks-dualizing/pinned-original-source.tex, source lines 3831–3831. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-depth-ext, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-depth: sources/stacks-dualizing/pinned-original-source.tex, source lines 1921–1921. Navigation pointer only: the counted text refers to the Stacks item lemma-depth, which is an existing labeled result of

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this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-dualizing: sources/stacks-dualizing/pinned-original-source.tex, source lines 2818–2818. Navigation pointer only: the counted text refers to the Stacks item lemma-dualizing, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-sitting-in-degrees: sources/stacks-dualizing/pinned-original-source.tex, source lines 3232–3232. Navigation pointer only: the counted text refers to the Stacks item lemma-sitting-in-degrees, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-special-case-local-duality: sources/stacks-dualizing/pinned-original-source.tex, source lines 3727–3727. Navigation pointer only: the counted text refers to the Stacks item lemma-special-case-local-duality, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a Noetherian local ring with normalized dualizing complex $\omega_A^\bullet$. Then depth(A) is equal to the smallest integer $\delta \geq 0$ such that $H^{-\delta}(\omega_A^\bullet) \neq 0$.*

Proof. This follows immediately from Lemma lemma-sitting-in-degrees. Here are two other ways to see that it is true.

First alternative. By Nakayama's lemma we see that δ is the smallest integer such that $\mathrm{Hom}_A(H^{-\delta}(\omega_A^\bullet), \kappa) \neq 0$. In other words, it is the smallest integer such that $\mathrm{Ext}_A^{-\delta}(\omega_A^\bullet, \kappa)$ is nonzero. Using Lemma lemma-dualizing and the fact that $\omega_A^\bullet$ is normalized this is equal to the smallest integer such that $\mathrm{Ext}_A^\delta(\kappa, A)$ is nonzero. This is equal to the depth of A by Algebra, Lemma algebra-lemma-depth-ext.

Second alternative. By the local duality theorem (in the form of Lemma lemma-special-case-local-duality) δ is the smallest integer such that $H^\delta_{\mathfrak{m}}(A)$ is nonzero. This is equal to the depth of A by Lemma lemma-depth. □